# FixoraAI

Enterprise Multimodal Brand Governance & Content Publishing Platform

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| **Hackathon Stage:** Round 3 Final Submission (Top 50) | **Team Leader:** Pranamya Kalambe |
| **Live Web Application:** https://fixoraai.onrender.com | **Production API Docs:** https://fixoraai-backend.onrender.com/docs |
| **Source Repository:** https://github.com/Pranamya1/FixoraAI | **Evaluator Access:** 1-Click Judge Access (No sign-up needed) |

## Executive Overview

Modern marketing and engineering organizations face a critical governance dilemma: generative AI enables teams to produce content 100× faster, but leaves enterprise brands vulnerable to severe brand dilution, non-compliant color palettes, visual design errors, and legal claims. **FixoraAI** solves this by embedding an intelligent, multimodal governance layer directly between creative generation and social publishing.

Rather than functioning as a surface-level LLM wrapper, FixoraAI extracts and indexes company brand guidelines into vector embeddings, performs multimodal visual inspection using **Google Gemini Vision** on creative posters, routes approval requests to executives across four channels (WhatsApp, Gmail, Slack, Teams), and executes 1-click live publishing via real OAuth.

## 1. Delivered Round 3 Improvements (Committed at Round 2)

As committed in the Round 2 milestone submission, all three designated major engineering improvements have been delivered, tested, and deployed to production:

| Delivered Improvement | Technical Implementation & User Value |
|---|---|
| **Improvement 1: Direct Multi-Platform Social Publishing** | • **Real LinkedIn OAuth 2.0:** Engineered live authorization code flow with secure token persistence in Supabase, enabling 1-click publishing directly to live LinkedIn feeds.<br>• **Multi-Channel Routing:** Added Twitter/X pre-formatted Web Intent dispatch and Slack channel webhook integration, eliminating manual copy-pasting for marketing teams. |
| **Improvement 2: Multimodal Creative Visual Auditing** | • **Gemini Vision Inspection:** Implemented deep visual inspection of generated posters and uploaded images to verify brand color hex tokens (e.g., #009DFF primary), typographical legibility, logo placement, and layout balance.<br>• **Dual Scorecards:** Provides creators with instant visual compliance ratings alongside textual guideline checks. |
| **Improvement 3: Executive Review & Radio Dispatch Hub** | • **Single-Platform Radio Selector:** Redesigned dispatch modal to single-channel radio selection, preventing multi-channel notification spam.<br>• **Cross-Platform Gateways:** Instant WhatsApp wa.me mobile chat links, executive responsive HTML emails, interactive Slack Block Kit cards, and Microsoft Teams Connector cards with custom webhook overrides. |

## 2. Complete Production Feature Matrix

| Category | Feature Name | Capabilities & Architectural Details | Status |
|---|---|---|---|
| **Knowledge Ingestion** | Brand Guideline Vector Parser | Extracts text and layout from enterprise PDFs/Markdown; chunks and generates 768-dim embeddings in Supabase pgvector. | **Production** |
| **Compliance Engine** | RAG Semantic Auditor | Audits drafts against indexed company policies with rule citations, severity classifications (Critical/Warning/Notice), and 1-click fixes. | **Production** |
| **Multimodal AI** | Gemini Vision Visual Inspector | Visual inspection of graphic layouts, contrast ratios, and color palette tokens (#009DFF) powered by Gemini Vision. | **Production** |
| **Creative Studio** | Multi-Deliverable Synthesis | Generates channel-calibrated copy, 1080×1080 branded social posters (.PNG via Pillow), and 4-slide PDF carousels (ReportLab). | **Production** |
| **Review & Sign-Off** | Executive Approval Dispatch | Single-platform radio dispatch across WhatsApp, Gmail HTML cards, Slack Block Kit, and Microsoft Teams cards. | **Production** |
| **Social Publishing** | Direct Channel Integration | Live OAuth 2.0 direct feed posting to LinkedIn; verified Twitter/X web intent; channel handle linking in settings. | **Production** |
| **Authentication** | 1-Click Judge Access | Instant evaluator walkthrough button bypassing onboarding and credential friction while supporting full Supabase Auth. | **Production** |

## 3. Technical Architecture & Implementation Summary

FixoraAI is engineered as a decoupled, high-throughput full-stack system built for low-latency governance verification:

- **Frontend Client:** React 19, TypeScript, and Vite styled with modern Tailwind CSS v4. State persistence is managed through resilient local storage synchronization and reactive event listeners.
- **Backend API:** Python 3.10+ with FastAPI utilizing asynchronous request handling, Pydantic data contract validation, and connection pooling for external services.
- **AI & Multimodal Core:** Google Gemini API (gemini-3-flash-preview for contextual reasoning and structured audit outputs; Gemini Vision for graphic and typography validation; models/embedding-001 for semantic retrieval).
- **Database & Vector Store:** Supabase PostgreSQL enhanced with the pgvector extension for sub-50ms cosine similarity searches against chunked brand guidelines.
- **Asset Synthesis Pipelines:** Headless Python graphics pipelines utilizing Pillow (PIL) for crisp 1080×1080 brand poster composition and ReportLab for structured, multi-page LinkedIn carousel PDFs.
- **Enterprise Integrations:** Native OAuth 2.0 protocol for LinkedIn live publishing, Webhook dispatchers for Slack/Teams, and dynamic deep-linking for WhatsApp and mailto clients.

## 4. Reliability & Independent Evaluator Testing

To satisfy the Round 3 requirement that *'a judge will open your link and use it on their own with nobody from your team explaining anything,'* FixoraAI incorporates comprehensive automated defensive engineering:

- **Zero-Setup Graceful Fallback:** If a judge begins generating content without uploading custom guidelines, FixoraAI automatically falls back to an enterprise default brand kit, preventing null-pointer errors or empty state

blockers.

- **SPA Cloud Routing Protection:** Implemented a production _redirects rewrite policy (/* /index.html 200) to ensure deep links, settings tabs (/?tab=settings), and OAuth redirect callbacks never produce 404 Not Found errors on Render.
- **State Resilience on Browser Refresh:** Active review sessions, generated drafts, and user tokens persist across hard browser reloads through atomic storage synchronization.
- **Defensive Input Validation:** All API routes enforce strict schema boundaries via Pydantic; invalid phone formats, malformed webhook URLs, and empty topics display clean inline warnings rather than throwing unhandled 500 exceptions.
- **Build & Type Integrity:** 100% clean TypeScript compilation check (npm run build, zero errors across 2,100+ transformed modules) and clean Python backend import verification.

## 5. Product Roadmap & Future Commercial Development

| Phase / Milestone | Key Deliverables & Business Impact |
|---|---|
| **Q1 2027**<br>**Enterprise SSO & RBAC** | Implement SAML 2.0 / Okta enterprise identity with hierarchical role-based permissions (Content Creator, Compliance Officer, Chief Brand Officer). |
| **Q2 2027**<br>**Video Brand Governance** | Extend multimodal inspection to short-form video (TikTok, Instagram Reels, YouTube Shorts) utilizing frame-by-frame audio and visual compliance scanning. |
| **Q3 2027**<br>**Automated Legal Localization** | Incorporate jurisdiction-aware legal compliance checks (FTC endorsement guidelines, GDPR cookie/privacy declarations, and ASCI advertising regulations). |
| **Q4 2027**<br>**Agency Multi-Brand Suite** | Develop a multi-tenant portal enabling marketing and PR agencies to govern and publish across hundreds of distinct client brand guidelines from a unified cockpit. |

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**Submitted by:** Pranamya Kalambe (Team Leader) | **Product:** FixoraAI | **Round:** 3 (Final)
**Production URL:** https://fixoraai.onrender.com | **Code Repository:** https://github.com/Pranamya1/FixoraAI